EM-Wave in a Polar Crystal

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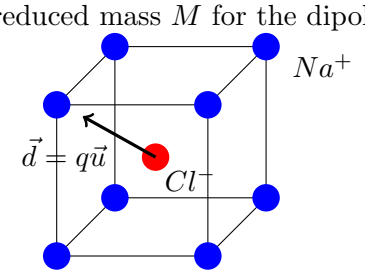
1 Problem

1.1 Set Up

For today's problem, we wish to consider EM-waves in a polar crystal such as $NaCl$. Here, the polarization arises from the ionic bonding between the Na and Cl atoms resulting in an electric dipole moment within the crystal lattice. This moment is given by $\vec{d} = q\vec{u}(\vec{r}, t)$, where $\vec{u}$ is the displacement vector of our system and q is the relevant charge in the dipole. We will assume that this system is coupled internally, between the atoms of the dipole, via a spring force with some natural frequency ω_0. We will also parametrize our crystal with a reduced mass M for the dipole.[1]



There are two scales at which the interaction happens: a microscopic scale in which we view the oscillation of the atoms in the crystal, and a macroscopic scale in which we only see the changing electromagnetic field. In the case of the microscopic, we immediately have from Newton II that

$$M\ddot{\vec{u}} = -k\vec{u} + q\vec{E} \tag{1}$$

with $k = M\omega_0^2$, and for the macroscopic from Maxwell's Equations that

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -i\omega\vec{B} \tag{2}$$

[1] We could also write down equations for each of the sodium and the chloride atoms where each gains its own displacement vector u_{Na} and u_{Cl}, respectively. Doing this results in two equations along the lines of Equation 1. Solving these two equations results an equation of the form of our Equation 1, so no physics has been hastily and unjustifiably swept under the rug.

where we have imposed that the E and B-fields are sinusoidal with frequency ω. We also know from Maxwell's equations that

$$\mu_0 \vec{\nabla} \times \vec{H} = \mu_0 \frac{\partial \vec{D}}{\partial t}.$$

Since $\vec{E}$ and $\vec{B}$ are sinusoidal it follows that $\vec{D}$ is also sinusoidal since the crystal's polarization is a direct response to the E-field. If we also note that $\vec{\nabla} \times \vec{H} = \vec{B}/\mu_0$ and use the (theoretical) definition of c, we get

$$\vec{\nabla} \times \vec{B} = -\frac{i\omega}{\epsilon_0 c^2} \vec{D}. \tag{3}$$

Taking the curl of Equation 2,

$$\begin{aligned}
\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) &= i\omega \vec{\nabla} \times \vec{B} \\
\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E} &= \frac{\omega^2}{\epsilon_0 c^2} \vec{D} \quad \text{by (3)} \\
&= \frac{\omega^2}{\epsilon_0 c^2} \left(\epsilon_0 \vec{E} + \frac{\vec{d}}{V} \right) \\
&= \frac{\omega^2}{\epsilon_0 c^2} \left(\epsilon_0 \vec{E} + qN\vec{u} \right)
\end{aligned}$$

The next to last step uses the definition of the electric displacement vector, noting that the polarization is just the dipole moment density (*i.e.* $\vec{P} = \vec{d}/V$ with the volume of the crystal's unit cell). The last step notes that we can equivalently write V in terms of the unit cell's number density (the number of cells per unit volume $N = V^{-1}$).

Equation 1 gives us the means to cast the above into a form solely in terms of $\vec{E}$. We can easily handle the derivative by noticing that, as with the electric displacement vector, our dipole's displacement vector will respond with and thus oscillate with the E-field.

$$\begin{aligned}
(1) \Rightarrow \vec{u} &= \frac{q}{k - \omega^2 M} \vec{E} \\
\Rightarrow \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E} &= \left(\frac{\omega^2}{c^2} + \frac{N\omega^2 q^2}{\epsilon_0 c^2 (k - \omega^2 M)} \right) \vec{E} \\
&= \left(\frac{\omega^2}{c^2} + \frac{\omega^2 \Omega^2}{c^2(\omega_0^2 - \omega^2)} \right) \vec{E}
\end{aligned} \tag{4}$$

The new parameter $\Omega^2 := q^2 N/\epsilon_0 M$ is reminiscent of the plasma frequency $\omega_p^2 = q^2 n_e/\epsilon_0 m_e$. The difference between the two is that instead of electrons, we now have ions; thus we call Ω the *ion plasma frequency.*

There are two cases that give solutions to our problem: An oscillating E-field that is either longitudinal or transverse. We begin with the former.

1.2 Longitudinal E-Field

For the longitudinal E-field, we let $\vec{E} = E_0\hat{k}e^{i\vec{k}\cdot\vec{r}}$. We are free to choose a convenient coordinate system so we may assume that $\hat{k} = \hat{x}$. Now the derivatives are much easier to see, each of which gives us a factor of k. The Lapacian naturally does not affect the direction of our vector while the gradient-divergence combination first gets rid of all the direction and then picks back up a factor of $\hat{x}$ since the only non-zero derivative is the x-component. So we have that:

$$\begin{aligned}
(4) \rightarrow \left(\frac{\omega^2}{c^2} + \frac{\omega^2\Omega^2}{c^2(\omega_0^2-\omega^2)}\right)\vec{E} &= k^2\vec{E} - k^2\vec{E} \\
\Rightarrow \left(\frac{\omega^2}{c^2} + \frac{\omega^2\Omega^2}{c^2(\omega_0^2-\omega^2)}\right) &= 0 \\
\frac{\Omega^2}{(\omega_0^2-\omega^2)} &= -1 \\
\therefore \omega(k) &= \sqrt{\omega_0^2+\Omega^2}. \qquad (5)
\end{aligned}$$

1.3 Transverse E-Field

We let the E-field have the same form as before, but fix its direction by imposing $\vec{k}\cdot\vec{E} = 0$. As before, the Laplacian gives us two factors of k, but now the $\vec{\nabla}\cdot\vec{E} = 0$. This follows from the fact that, letting $\vec{E} \perp \hat{k} = \hat{x}$; so the only non-zero derivative is in the x-direction, but this component dots into nabla to yield zero. In this case, our equation becomes:

$$\begin{aligned}
(4) \rightarrow k^2\vec{E} &= \left(\frac{\omega^2}{c^2} + \frac{\omega^2\Omega^2}{c^2(\omega_0^2-\omega^2)}\right)\vec{E} \\
\Rightarrow k^2 &= \frac{\omega^2}{c^2} + \frac{\omega^2\Omega^2}{c^2(\omega_0^2-\omega^2)} \\
(\omega^2-\omega_0^2)c^2k^2 &= (\omega^2-\omega_0^2)\omega^2 + \Omega^2\omega^2 \\
\omega^4 - (c^2k^2+\omega_0^2+\Omega^2)\omega^2 + c^2k^2\omega_0^2 &= 0.
\end{aligned}$$

Which is quadratic in ω^2. Applying the quadratic formula, we find two solutions:

$$\therefore \omega_\pm^2(k) = \frac{1}{2}\left(c^2k^2+\omega_0^2+\Omega^2 \pm \sqrt{(c^2k^2+\omega_0^2+\Omega^2)^2 - 4c^2k^2\omega_0^2}\right). \qquad (6)$$

1.4 Analysis

In order to graph and understand the behaviors of the dispersion relations for the longitudinal and transverse E-field cases, we first must find the limiting behavior of them. The longitudinal case has trivial behavior, so we focus on the dispersion relation for the transverse EM-wave.

$\mathbf{k} \to \mathbf{0}$
Taylor expanding Equation 6:

$$\omega_{\pm}^2(k) = \frac{1}{2}\left(\omega_0^2 + \Omega^2 \pm (\omega_0^2 + \Omega^2 + \frac{\omega_0^2 - \Omega^2}{\omega_0^2 + \Omega^2} c^2k^2 + \mathcal{O}(k^4))\right).$$

We thus get $\omega_+(k) \overset{k\to 0}{\sim} \sqrt{\omega_0^2 + \Omega^2 + c^2k^2}$ and $\omega_-(k) \overset{k\to 0}{\sim} ck$. For the ω_+ solution, the limit is dominated by the constant term which is also the intercept. For the ω_- solution, the dispersion relation tends to zero as a linear function of k.

$\mathbf{k} \to \infty$

$$\begin{aligned}
\omega_{\pm}^2(k) &= \frac{1}{2}\left(\omega_0^2 + \Omega^2 + c^2k^2 \pm c^2k^2\sqrt{\left(\left(\frac{\omega_0}{ck}\right)^2 + \left(\frac{\Omega}{ck}\right)^2 + 1\right)^2 - 4\left(\frac{\omega_0}{ck}\right)^2}\right) \\
&= \frac{1}{2}\left(\omega_0^2 + \Omega^2 + c^2k^2 \pm c^2k^2\sqrt{\left(\left(1 + \frac{\Omega^2}{\omega_0^2}\right)x^2 + 1\right)^2 - 4x^2}\right) \;:\; x = \frac{\omega_0}{ck} \ll 1 \\
&= \frac{1}{2}\left(\omega_0^2 + \Omega^2 + c^2k^2 \pm c^2k^2\left(1 + \left(\frac{\Omega^2}{\omega_0^2} - 1\right)x^2 + \mathcal{O}(x^4)\right)\right) \\
&\approx \frac{1}{2}\left(\omega_0^2 + \Omega^2 + c^2k^2 \pm (c^2k^2 + \Omega^2 - \omega_0^2)\right)
\end{aligned}$$

We can now see that, in the large k regime, $\omega_+ \sim ck$, and $\omega_- \sim \omega_0$.

The limits where $\omega(k)$ is linear in k with a slope c corresponds to a *photon-like* dispersion relation. As the frequency changes (decreasing for $\omega_+(k)$ and increasing for $\omega_-(k)$). The modes of our wave move to a *phonon-like* state. For interim frequencies, the modes are able to couple to one another (*i.e.* ω_+ with ω_-) and form an intermediate state that is neither photon nor phono-like. This state is a mixture of the two, and is known as a *polariton* mode.

The existence of the polariton mode depends on the existence of non-zero polarization in the crystal. As the charge forming the dipole (the variable q above) tends to zero, the two modes disappear leaving only the photon-like mode. This is to be expected as, with

the dipole disappearing, we are left with a crystal made out of atoms (as opposed to ions). In this case, we have effectively turned off the harmonic component in the system, and are left with only the influence of the incident EM-wave– a photon-like mode as it is just a photon propagating unperturbed through the crystal. Also worth noting is that, with $q \to 0$, the frequency gap of the EM-wave in our crystal disappears, and level crossing between the longitudinal and transverse solutions emerges.

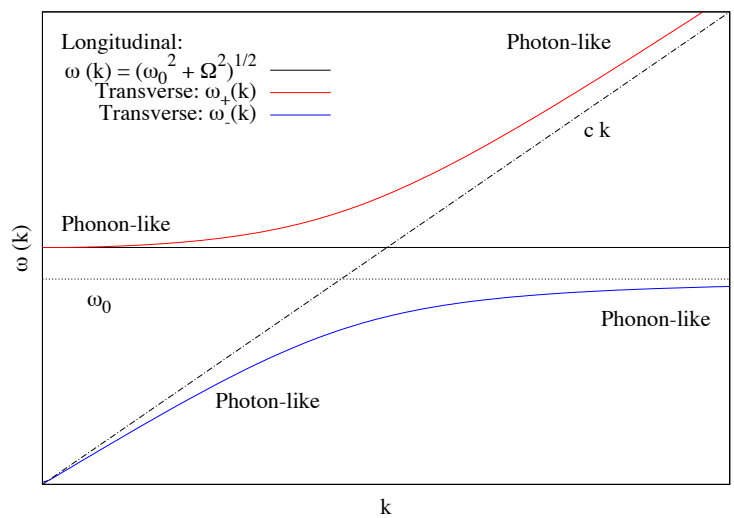


Figure 1: Dispersion relations for the transverse and longitudinal modes of an EM-wave in a polar crystal as found above. We can see that the ω_+ solution (red) tends to the phonon-like, longitudinal solution (black) at low wave numbers while becoming photon-like for high wave numbers. The opposite trend (in terms of wave numbers) is seen for for theω_- solution (blue) except that the phonon-like regime takes on frequencies asymptotic to the electron plasma frequency, ω_0, and not the longitudinal solution's frequency. These limits as well as the purely photon-like dispersion relation ($\omega = ck$) are plotted for reference.

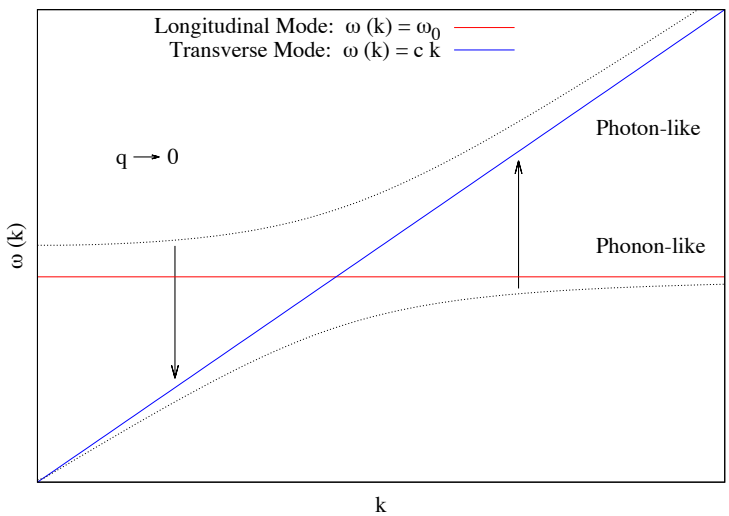


Figure 2: The affect of letting $q \to 0$ on the dispersion relations. As the polar nature of the crystal is reduced, the transverse modes found above (grey dashed) tend to the pure photon-like state (blue) as demonstrated by the arrows. Note that the longitudinal solution (red) has shifted downwards to the electron plasma frequency, ω_0, as the ion plasma frequency, Ω, goes to 0 with q.